

Practical 10

Prac(a) 10:

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polisettysreeharsha@SREEHARSHA:~$ echo "This is inode practical." > original.txt
polisettysreeharsha@SREEHARSHA:~$ cat original.txt
This is inode practical.
polisettysreeharsha@SREEHARSHA:~$ ls -li original.txt
22327 original.txt
polisettysreeharsha@SREEHARSHA:~$ stat original.txt
  File: original.txt
  size: 25          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 22327       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/polisettysreeharsha)   Gid: ( 1000/polisettysreeharsha)
Access: 2026-10-06 15:32:16.562801235 +0000
Modify: 2026-10-06 15:31:13.894804171 +0000
Change: 2026-10-06 15:31:13.894804171 +0000
 Birth: 2026-10-06 15:31:13.894804171 +0000
polisettysreeharsha@SREEHARSHA:~$ ln original.txt hardlink.txt
polisettysreeharsha@SREEHARSHA:~$ ls -li original.txt hardlink.txt
22327 hardlink.txt 22327 original.txt
polisettysreeharsha@SREEHARSHA:~$ stat original.txt
  File: original.txt
  size: 25          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 22327       Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/polisettysreeharsha)   Gid: ( 1000/polisettysreeharsha)
Access: 2026-10-06 15:32:16.562801235 +0000
Modify: 2026-10-06 15:31:13.894804171 +0000
Change: 2026-10-06 15:32:58.346798897 +0000
 Birth: 2026-10-06 15:31:13.894804171 +0000
polisettysreeharsha@SREEHARSHA:~$ ln -s original.txt symlink.txt
polisettysreeharsha@SREEHARSHA:~$ ls -li original.txt hardlink.txt symlink.txt
22327 -rw-r--r-- 2 polisettysreeharsha polisettysreeharsha 25 Oct  6 15:31 hardlink.txt
22327 -rw-r--r-- 2 polisettysreeharsha polisettysreeharsha 25 Oct  6 15:31 original.txt
26368 lrwxrwxrwx 1 polisettysreeharsha polisettysreeharsha 12 Oct  6 15:33 symlink.txt -> original.txt
polisettysreeharsha@SREEHARSHA:~$ ls -li original.txt
22327 original.txt
polisettysreeharsha@SREEHARSHA:~$ find . -inum 22327
./hardlink.txt
./original.txt
polisettysreeharsha@SREEHARSHA:~$ echo "New Line added." >> hardlink.txt
polisettysreeharsha@SREEHARSHA:~$ cat original.txt
This is inode practical.
New line added.
polisettysreeharsha@SREEHARSHA:~$ cat symlink.txt
This is inode practical.
New line added.
polisettysreeharsha@SREEHARSHA:~$ |
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Prac(b) 10:

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polisettysreeharsha@SREEHARSHA:~$ nano Prac10bcode.c
polisettysreeharsha@SREEHARSHA:~$ echo "This is a test file for mmap practical." > mmap_input.txt
polisettysreeharsha@SREEHARSHA:~$ cat mmap_input.txt
This is a test file for mmap practical.
polisettysreeharsha@SREEHARSHA:~$ gcc Prac10bcode.c -o Prac10bcode
polisettysreeharsha@SREEHARSHA:~$ ./Prac10bcode
Memory-Mapped I/O vs Traditional I/O
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Traditional read/write copy completed.
Time taken: 0.000120 seconds

Memory-mapped copy completed.
Time taken: 0.010822 seconds

Performance Comparison:
Traditional read/write : 0.000120 seconds
mmap()                  : 0.010822 seconds

Files created successfully:
- copy_read_write.txt
- copy_mmap.txt
polisettysreeharsha@SREEHARSHA:~$ cat copy_read_write.txt
This is a test file for mmap practical.
polisettysreeharsha@SREEHARSHA:~$ cat copy_mmap.txt
This is a test file for mmap practical.
polisettysreeharsha@SREEHARSHA:~$ cmp copy_read_write.txt copy_mmap.txt
polisettysreeharsha@SREEHARSHA:~$ |
```